

SEATTLE HARBOR, WA, USA

WMO: 994014

Lat: 47.6026N

Lon: 122.3392W

Elev: 2

StdP: 101.30

Time Zone: -8.00 (NAP)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -1.7	(c) 0.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 9.1	(k) 9.9	(l) 7.5	(m) 8.2	(n) 2.0	(o) 130	(p) 0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.2	(c) 26.3	(d) N/A	(e) 24.4	(f) N/A	(g) 22.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.9	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.3	(b) 6.0	(c) 5.2	(e) -4.4	(f) 30.6	(g) 2.0	(h) 2.2	(i) -5.8	(j) 32.2	(k) -6.9	(l) 33.5	(m) -8.0	(n) 34.8	(o) -9.5	(p) 36.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	6.7	7.1	8.6	10.5	13.6	15.8	18.3	18.3	16.4	12.6	8.6	6.2
	DBStd	5.02	2.81	2.71	2.42	2.30	2.56	2.32	2.14	1.76	2.09	2.17	2.97	3.14
	HDD10.0	457	108	84	56	19	1	0	0	0	0	4	61	123
	HDD18.3	2403	362	315	300	235	148	82	27	22	64	177	293	377
	CDD10.0	1158	4	3	14	34	113	174	257	258	192	86	17	4
	CDD18.3	63	0	0	0	0	2	7	26	22	7	0	0	0
	CDH23.3	280	0	0	0	1	19	41	121	81	18	0	0	0
	CDH26.7	45	0	0	0	0	4	6	24	10	2	0	0	0
Wind	WSAvg	2.2	2.3	2.3	2.7	2.6	2.3	2.2	2.0	1.8	1.9	2.0	2.4	2.2
Precipitation	PrecAvg	985	133	101	100	73	47	41	25	32	47	84	143	157
	PrecMax	1384	264	219	222	147	108	103	108	139	143	204	284	304
	PrecMin	706	15	14	11	22	11	3	0	0	0	7	13	31
	PrecStd	153	59	46	44	30	23	25	20	31	35	47	56	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	14.5	17.7	20.9	25.6	27.3	29.7	27.8	25.5	20.3	16.5	15.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	12.5	12.8	14.8	17.6	21.9	24.3	26.5	25.5	23.0	17.9	14.6	12.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	11.3	11.5	13.3	15.4	19.3	21.6	24.4	23.7	21.2	16.8	13.7	11.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	10.2	10.5	12.1	13.9	17.6	19.8	22.5	22.0	19.8	15.7	12.6	10.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.5	5.6	6.4	6.9	7.0	8.2	8.0	6.4	4.8	4.4	3.9
		MCDBR	4.7	5.9	7.2	8.9	10.3	10.3	11.7	10.9	9.1	6.3	4.6	4.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.328	0.318	0.327	0.358	0.377	0.366	0.353	0.359	0.347	0.332	0.323	0.332
	taud		2.488	2.548	2.541	2.426	2.392	2.447	2.503	2.495	2.519	2.564	2.539	2.410
	Ebn at Noon		753	856	900	896	886	897	905	885	865	826	757	689
	Edh at Noon		65	74	87	107	115	109	102	99	88	72	61	62
All-Sky Solar Radiation	RadAvg		1.01	1.88	2.84	4.18	5.27	5.78	6.46	5.50	3.99	2.25	1.18	0.82
	RadStd		0.14	0.25	0.33	0.39	0.43	0.57	0.48	0.36	0.36	0.25	0.16	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page